GROUP WORK PROJECT # 2 MScFE 560: FINANCIAL MARKETS
GROUP NUMBER: 5805

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Overview

1. An open-ended, low-cost mutual fund that tracks a broad market index:

1.1. Introduction

An open-ended, low-cost mutual fund that tracks a broad market index is created to imitate the performance of a comprehensive market index, like the S&P 500 or the total stock market index. These funds offer an easy and reasonably priced option for investors to obtain exposure to a diverse portfolio of stocks, bonds, and other securities.

1.2. Performance

- Market Correlation: These funds aim to replicate the returns of the index they track, historically providing solid long-term growth. For example, the S&P 500 has averaged an annual return of approximately 10% over the past several decades. [1]

- Reduced Risk through Diversification: By investing in a wide array of companies across various sectors, these funds mitigate the risk associated with individual stock volatility. This diversification helps stabilise returns over the long term.

- Long-Term Growth: These funds are well-positioned for steady appreciation because they include many of the largest and most successful companies.

1.3. Fees

- Expense Ratios: The expense ratios for these funds are typically much lower than those of actively managed funds. For instance, many index funds have expense ratios below 0.10%, compared to 1% or more for actively managed funds.

- No Load Fees: Many of these funds are no-load, meaning there are no sales commissions or fees when purchasing or redeeming shares, further reducing the cost to investors.

- Low Turnover Costs: The passive management style leads to lower portfolio turnover, which minimizes transaction costs and tax implications.

1.4. Transparency

- Simple Strategy: TrackingTracking a broad market index is straightforward and easy for investors to understand. The fund's objective and holdings are clear and predictable.

- Regular Reporting: These funds provide regular and detailed reports, including quarterly and annual reports, showing the fund's performance, holdings, and expense information. This allows investors to stay informed about where their money is invested.

-Daily NAV Calculation: The fund's net asset value (NAV) is calculated daily and published, providing transparency in pricing.

1.5. Liquidity

- Daily Liquidity: Investors can buy or sell shares at the fund's NAV at the end of each trading day. This provides ample liquidity for those needing access to their money.

-Redemption Flexibility: Unlike some other investment vehicles, there are typically no restrictions or penalties for selling shares, making these funds a flexible option for investors.

1.6. Professional Management

- Index Replication: Professional fund managers are responsible for maintaining a portfolio that closely tracks the index's performance. This involves selecting and weighing the appropriate securities to match the index.

- Portfolio Rebalancing: Managers handle periodic rebalancing of the fund's holdings to ensure alignment with the index, especially after index reconstitutions or corporate actions.

- Operational Efficiency: The management team ensures operational efficiency to keep tracking errors—the deviation of the fund's performance from the index's performance—as low as possible.

1.7. Investor Protections

These funds are subject to strict regulatory oversight from organisations such as the U.S. Securities and Exchange Commission (SEC), ensuring adherence to disclosure and reporting standards to protect investors and maintain fair practices. Investors also benefit from limited protection through the Securities Investor Protection Corporation (SIPC), which covers up to $500,000 in securities and cash in the event of brokerage firm failure, with a $250,000 limit for cash holdings. Fund managers have a fiduciary duty to prioritise investors' interests over their own, ensuring responsible fund management. Additionally, the Financial Industry Regulatory Authority (FINRA) oversees mutual funds, promoting fair and ethical practices in their marketing and sale processes to bolster investor confidence in these investment vehicles.

2. SPDR S&P 500 ETF Trust (SPY):

2.1. Introduction

The SPDR S&P 500 ETF Trust (SPY)[1], introduced in 1993, is not only the first ETF listed in the US but also the largest exchange-traded fund by net assets in the United States. This ETF is designed to follow the S&P 500® Index (the "Index") before expenditures[2]. SPY is a pivotal vehicle for investors looking to gain comprehensive exposure to the U.S. equity market, encompassing all eleven GICS sectors[3].

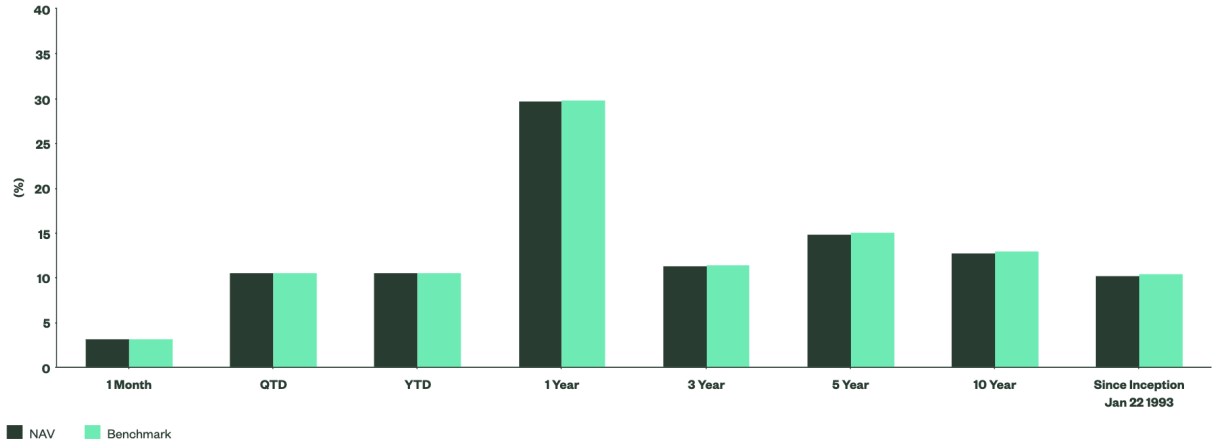
2.2. Performance

Overview | Performance | Holdings | Documents | Purchase Information | Morningstar Fund Comparison

Month End | **Quarter End**

Fund Inception Date: Jan 22 1993

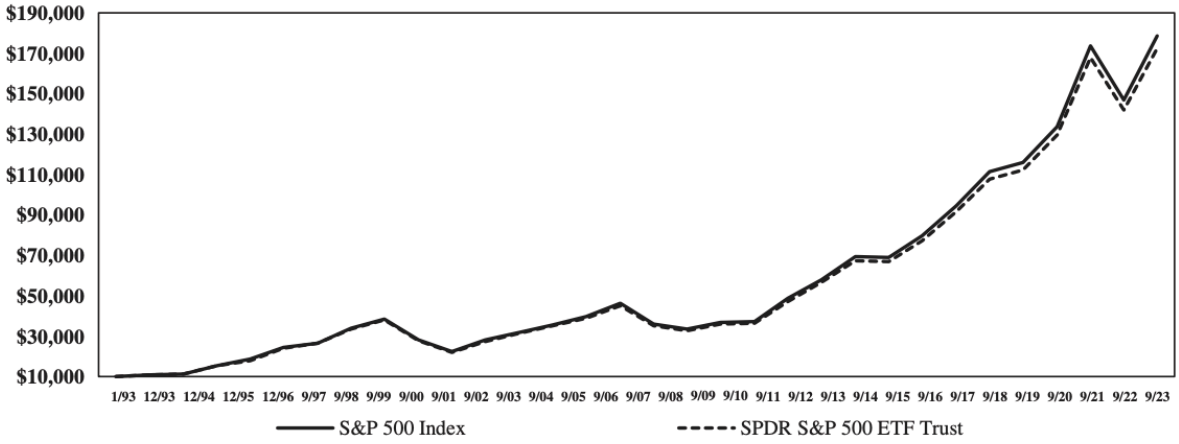
as of Mar 31 2024



	As Of	1 Month	QTD	YTD	1 Year	3 Year	5 Year	10 Year	Since Inception Jan 22 1993
Fund Before Tax									
NAV	Mar 31 2024	3.21%	10.53%	10.53%	29.71%	11.36%	14.89%	12.82%	10.30%
Market Value	Mar 31 2024	3.30%	10.40%	10.40%	29.65%	11.30%	14.88%	12.81%	10.30%
Benchmark ⓘ	Mar 31 2024	3.22%	10.56%	10.56%	29.88%	11.49%	15.05%	12.96%	10.43%
Fund After Tax									
Return after Taxes on Distributions	Mar 31 2024	3.08%	10.38%	10.38%	29.18%	10.95%	14.45%	12.33%	9.77%
Return after Taxes on Distributions and Sale of Fund Shares	Mar 31 2024	1.90%	6.23%	6.23%	17.76%	8.77%	11.89%	10.56%	9.00%
Benchmark ⓘ	Mar 31 2024	3.22%	10.56%	10.56%	29.88%	11.49%	15.05%	12.96%	10.43%

(Figure from SPDR® S&P 500® ETF Trust[1])

Growth of $10,000 Investment Since Inception[(1)(2)]



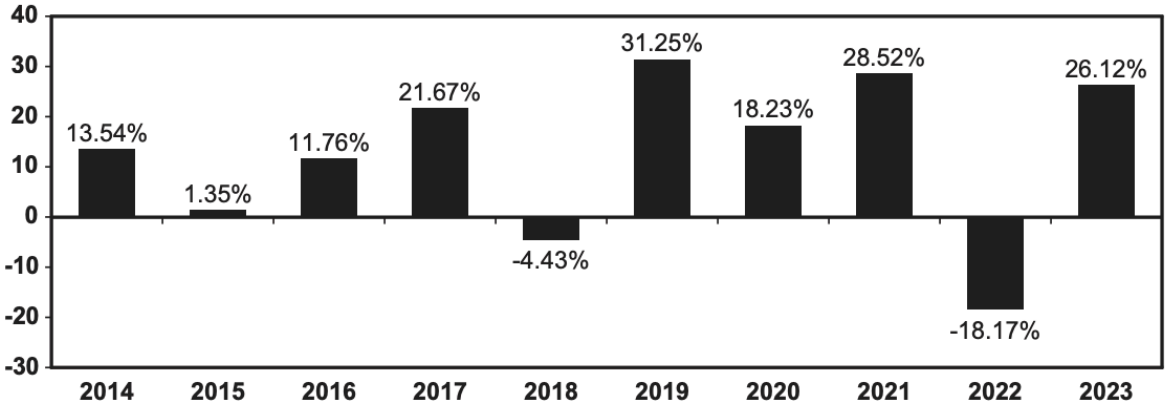
(1) Past performance is not necessarily an indication of how the Trust will perform in the future.

(2) Effective as of September 30, 1997, the Trust's fiscal year end changed from December 31 to September 30.

(Figure from SUMMARY section of SPDR® S&P 500® ETF Trust[4])

The SPDR S&P 500 ETF Trust (SPY) has not only well captured a record of performance but also consistently demonstrated significant long-term growth and recovery capabilities.

Annual Total Return (years ended 12/31)



(Figure from SUMMARY section of SPDR® S&P 500® ETF Trust[4])

Despite experiencing a drop in 2022 with a return of -18.17%, SPY has bounced back, underscoring its resilience and potential for growth over time. Additionally, its highest and lowest quarterly returns in recent years—20.44% in Q2 2020 and -19.60% in Q1 2020—highlight the fund's volatility and swift recovery in fluctuating market conditions[5].

2.3. Fees

One of SPY's key advantages is its exceptionally low **Gross Expense Ratio of 0.0945%**.
Ex: An annual management fee of just $9.45 for every $10,000 invested
This makes it a cost-effective vehicle for gaining broad market exposure. Low fees are very crucial for all investors as they directly impact net returns, and SPY's low expense ratio positions it as an advantageous option for long-term investors who want to minimize costs and maximize potential returns.

Unitholder Fees: None
(fees paid directly from your investment)

Estimated Annual Trust Ordinary Operating Expenses:
(expenses that you pay each year as a percentage of the value of your investment)

Current Estimated Annual Trust Ordinary Operating Expenses	As a % of Trust Average Net Assets
Trustee's Fee	0.0457%
S&P License Fee	0.0302%
Marketing	0.0172%
Other Operating Expenses	0.0015%
Total	**0.0945%***

* Does not sum due to rounding.

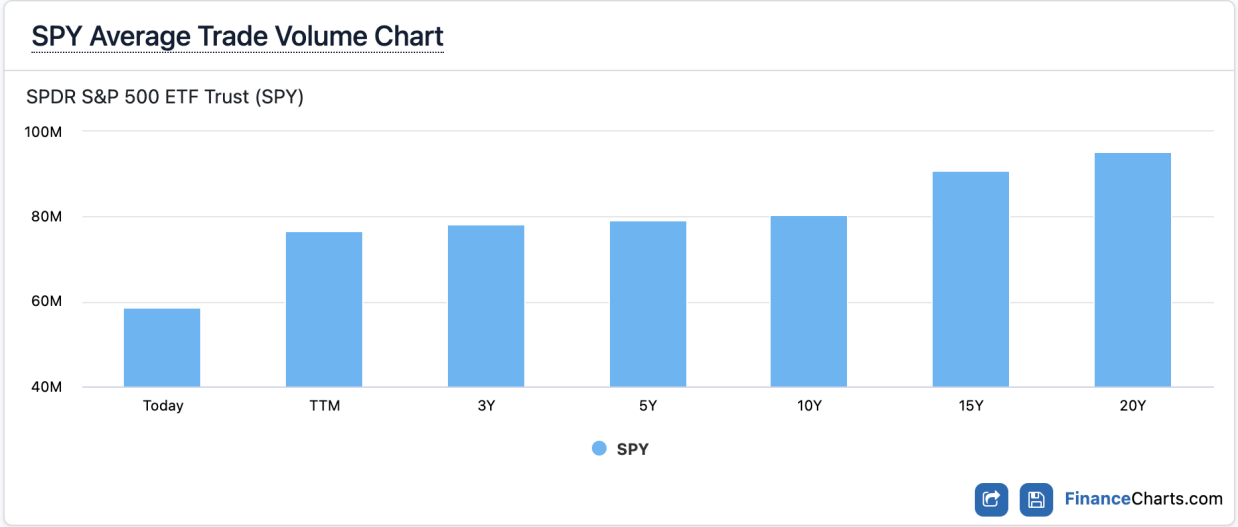
(Figure from SUMMARY section of SPDR® S&P 500® ETF Trust[4])

2.4. Transparency

SPY is well-known for its high level of transparency, which offers daily updates on its holdings, performance, and other vital fund information.
-> This transparency enables investors to see exactly what securities they own and make informed decisions based on the ETF's underlying assets. Regular updates and accessible information contribute to investor confidence and trust.

2.5. Liquidity

Also, SPY is the most well-established and liquid fund in the stock market. As the most liquid ETF, SPY benefits from high trading volumes, with an average daily trading volume consistently over 60 million shares. This liquidity is evidenced by the significant trading volumes observed over various time frames, including 3-year, 5-year, 10-year, 15-year, and 20-year periods.



(Figure from financecharts.com[17])

2.6. Professional Management

Managed by State Street Global Advisors, from there describe are "State Street Global Advisors Funds Distributors, LLC, member FINRA, SIPC, an indirect wholly owned subsidiary of State Street Corporation"

(Figure from State Street Global Advisors[18])

SPY benefits from the top-tier expertise of seasoned investment professionals dedicated to replicating and managing the fund's portfolio. While SPY is a passively managed fund, the professional oversight ensures that the ETF accurately tracks the S&P 500 Index. The fund's management involves rebalancing the portfolio to match the index and handling corporate actions like stock splits and dividends, ensuring that SPY remains a reliable tool for tracking the S&P 500.

2.7. Investor Protections

Although this is not FDIC, no Bank Guarantee and may lose value, investors in SPY benefit from several protections under U.S. financial regulatory frameworks. SPY is regulated by the Securities and Exchange Commission (SEC), ensuring compliance with stringent regulatory standards. Additionally, investments in SPY are protected by the Securities Investor Protection

Corporation (SIPC), which provides limited protection against broker-dealer insolvency. Furthermore, SPY operates under the fiduciary responsibilities of its trustees, who are obligated to act in the best interests of the investors. Oversight by FINRA (Financial Industry Regulatory Authority) also helps maintain market integrity and investor confidence.

2.8. Conclusion

The SPDR S&P 500 ETF Trust (SPY) stands out as a good choice for investors aiming to gain exposure to the S&P 500 Index. Its strong performance, low fees, high transparency, and liquidity, coupled with professional management and comprehensive investor protections, make it a compelling choice for both individual and institutional investors.

3. AN INDEXED ANNUITY

3.1. Introduction

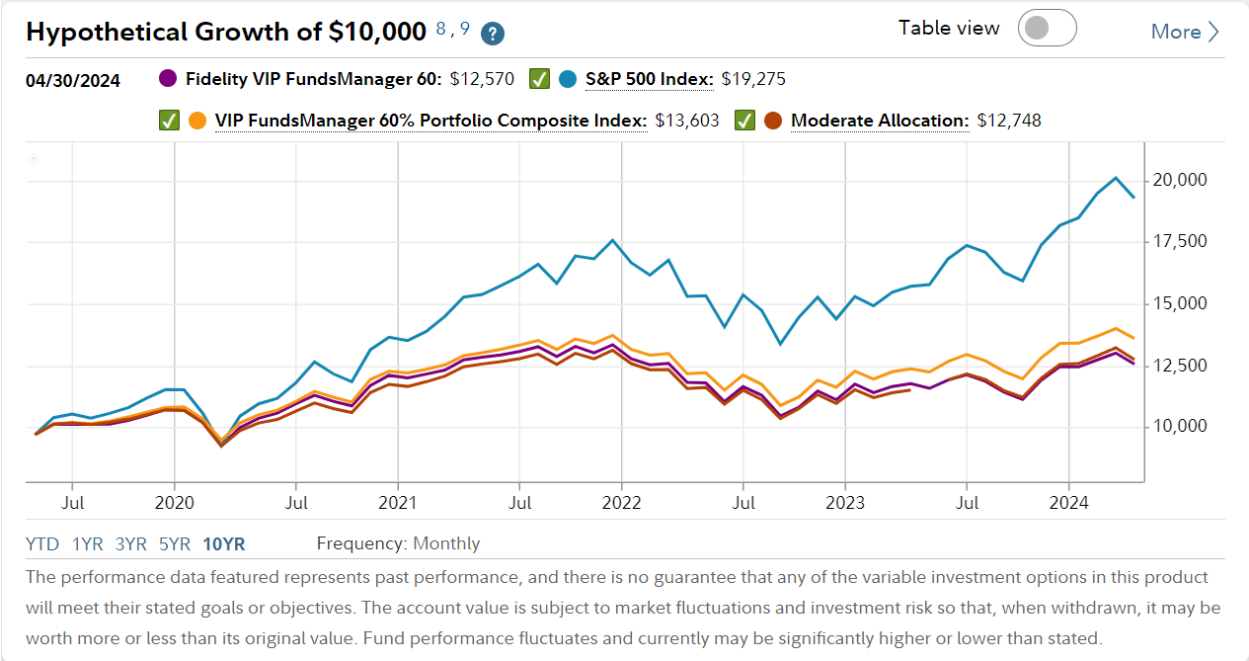
The *New York Life IndexFlex Variable Annuity - FP Series* is a single premium, deferred variable annuity designed to offer a combination of growth potential, downside protection, and tax-deferred savings for retirement or other long-term goals[1]. New York Life IndexFlex Variable Annuity- FP Series (IndexFlex) is an indexed annuity, it offers growth potential linked to a market index (S&P 500) while providing some level of protection against downturns.

3.2. Performance

New York Life IndexFlex Variable Annuity & New York Life IndexFlex Variable Annuity - FP Series Performance Summary
Average Annual Total Return as of February 29, 2024 (Updated Monthly)

				Non-Standardized (Assuming No Surrender)[2]							Standardized (Assuming Surrender)				
Investment Divisions[1]	Volatility/ Risk Level[3]	Portfolio Inception Date	Investment Division Inception Date[1]	Month to Date*	Year to Date* (Updated Monthly)	1 Year[4]	3 Year[4]	5 Year[4]	10 Year[4]	Since Investment Division Inception	1 Year[4]	3 Year[4]	5 Year[4]	10 Year[4]	Since Investment Division Inception
Equity															
MainStay VP S&P 500 Index – Service Class[5]	High	06/02/2003	01/19/2021	5.20%	6.80%	28.28%	10.06%	12.86%	10.76%	9.79%	20.28%	8.10%	12.23%	10.76%	8.18%
American Funds IS Growth Fund - Class 4	High	12/14/2012	05/01/2022	7.50%	9.22%	37.28%	6.67%	16.37%	13.35%	13.91%	29.28%	4.58%	15.82%	13.35%	9.93%
American Funds IS Washington Mutual Investors Fund - Class 4	High	12/14/2012	05/01/2022	4.40%	5.24%	22.78%	9.74%	10.37%	8.66%	10.09%	14.78%	7.77%	9.68%	8.66%	5.99%
Fidelity® VIP International Index Portfolio — Service Class 2	High	04/16/2018	01/19/2021	2.88%	1.11%	11.41%	-0.07%	3.96%	N/A	-0.66%	3.41%	-2.46%	3.09%	N/A	-2.62%
Franklin Templeton Aggressive Model Portfolio - Class II	High	05/01/2020	01/19/2021	4.59%	5.10%	19.91%	5.42%	N/A	N/A	4.97%	11.91%	3.28%	N/A	N/A	3.20%
Franklin Templeton Moderately Aggressive Model Portfolio - Class II	Moderate	05/01/2020	01/19/2021	3.12%	3.51%	15.94%	3.28%	N/A	N/A	2.87%	7.94%	1.04%	N/A	N/A	1.02%
Equity and Income															
MainStay VP Income Builder – Service Class	Moderate	06/02/2003	01/19/2021	1.47%	1.86%	9.56%	0.87%	3.65%	3.43%	0.38%	1.56%	-1.47%	2.76%	3.43%	-1.57%
American Funds IS Asset Allocation Fund - Class 4	Moderate	12/14/2012	01/19/2021	2.78%	3.63%	15.93%	3.62%	6.71%	5.96%	3.40%	7.93%	1.40%	5.93%	5.96%	1.57%
BlackRock® Global Allocation V.I. Fund – Class III Shares	Moderate	11/18/2003	01/19/2021	2.66%	2.46%	12.31%	-0.53%	5.25%	3.46%	-0.73%	4.31%	-2.91%	4.42%	3.46%	-2.68%
Fidelity® VIP FundsManager® 60% Portfolio — Service Class	Moderate	08/22/2007	01/19/2021	2.32%	2.30%	12.29%	2.17%	6.17%	4.83%	1.85%	4.29%	-0.11%	5.37%	4.83%	-0.04%
Franklin Templeton Moderate Model Portfolio - Class II	Moderate	05/01/2020	01/19/2021	2.42%	2.56%	12.87%	2.19%	N/A	N/A	1.79%	4.87%	-0.09%	N/A	N/A	-0.10%
Franklin Templeton Moderately Conservative Model Portfolio - Class II	Moderate	05/01/2020	01/19/2021	1.38%	1.27%	9.30%	0.63%	N/A	N/A	0.25%	1.30%	-1.73%	N/A	N/A	-1.70%
MainStay VP Janus Henderson Balanced – Service Class	Moderate	02/17/2012	01/19/2021	2.94%	4.75%	16.69%	4.40%	7.66%	6.73%	4.21%	8.69%	2.21%	6.91%	6.73%	2.41%
Income															
Fidelity® VIP Bond Index Portfolio – Service Class 2	Moderate	04/19/2018	01/19/2021	-1.54%	-1.75%	1.71%	-4.77%	-1.14%	N/A	-5.17%	-6.29%	-7.05%	-2.15%	N/A	-7.03%
Franklin Templeton Conservative Model Portfolio - Class II	Moderate	05/01/2020	01/19/2021	0.21%	0.19%	6.31%	-1.46%	N/A	N/A	-1.71%	-1.69%	-3.81%	N/A	N/A	-3.64%

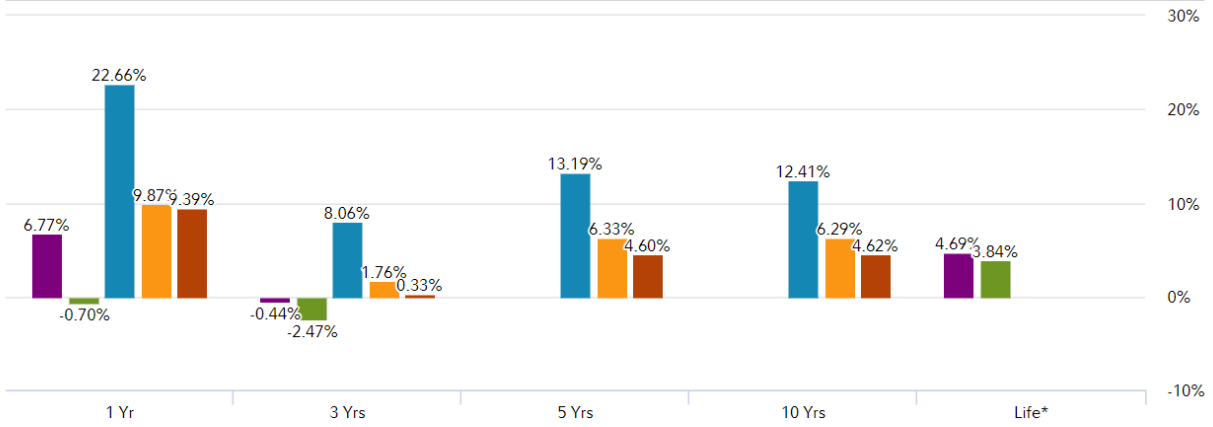
(Figure from the New York Life Performance Summary: IndexFlex Variable Annuity[2])



(Figure from the Fidelity VIP FundsManger 60[3])

Average Annual Total Returns 1, 2, 3, 4, 5, 6

AS OF 04/30/2024 | Morningstar Category:Moderate Allocation | Subaccount Inception 05/06/2019



	1 Yr	3 Yrs	5 Yrs	10 Yrs	Life*
FUND **Fidelity VIP FundsManager 60**	6.77%	-0.44%	--	--	4.69%
FUND **Fidelity VIP FundsManager 60 (After Surrender Fee)**	-0.70%	-2.47%	--	--	3.84%
PRIMARY BENCHMARK **S&P 500 Index**	22.66%	8.06%	13.19%	12.41%	--
SECONDARY BENCHMARK **VIP FundsManager 60% Portfolio Composite Index**	9.87%	1.76%	6.33%	6.29%	--
MORNINGSTAR CATEGORY AVERAGE **Moderate Allocation**	9.39%	0.33%	4.60%	4.62%	--
Rank in Morningstar Category	83%	70%	44%	56%	--
# of Funds in Morningstar Category	10,477	9,993	9,014	5,998	--

(Figure from the Fidelity VIP FundsManger 60[3])

3.3. Fees

The minimum premium is $10,000, and the Maximum premium exceeding $1 million requires NYLIAC approval.

Annual mortality and expense (M&E) and administrative charges are assessed only on amounts invested in the variable investment options and vary depending on the length of the surrender charge period.

Surrender-charge period:

5 Year	1.30%
6 Year	1.25%
7 Year	1.20%

In each policy year during the surrender charge period, the client may withdraw up to 10% of the account value as of the last policy anniversary. Withdrawals in excess of that amount are

subject to surrender charges:

Surrender-charge period:

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
5 Year	8%	8%	7%	6%	5%		
6 Year	8%	8%	7%	6%	5%	4%	
7 Year	8%	8%	7%	6%	5%	4%	3%

3.4. Transparency

The *New York Life IndexFlex Variable Annuity - FP Series* is a single-premium, deferred variable annuity with index-linked, fixed, and variable investment options. IndexFlex chooses from a carefully selected, streamlined menu of variable investment options from well-known asset managers, including model portfolios, as well as a fixed account. IndexFlex uses well-known indices S&P 500 for the index-linked option - no opaque or custom indices. Death benefit comes standard, at no additional cost.
IndexFlex offers index-linked investment options with interest credits based on the performance of well-established equity indices: S&P 500. For unlimited upside potential without the principal protection of the index-linked account, investors may allocate up to 100% of their investment across a variety of variable investment options from well-known money managers.

3.5. Liquidity

The New York Life IndexFlex Variable Annuity - FP Series offers some access to your money, but it has limitations that affect how "liquid" it is.

Features that reduce liquidity:

- **Surrender charges:** These are fees applied if you withdraw money within a certain period (usually up to 7 years) from buying into the annuity. They can significantly reduce your accessible cash.
- **Segment withdrawals:** If you withdraw money from an indexed investment option (index-linked account) before the end of its term (segment maturity date), you forfeit any potential growth from that period.

Features that offer some liquidity:

- **Partial withdrawals:** The policy allows you to take out some of your money after the initial surrender charge period.

- **Tax implications:** Early withdrawals (before age 59.5) may be subject to income tax and a 10% penalty.

3.6. Professional Management

The "Professional Management" aspect of the New York Life IndexFlex Variable Annuity - FP Series is actually self-directed, meaning you choose how your money is invested among the available variable investment options. There isn't a professional actively managing your account within the annuity itself.

3.7. Investor Protections

The New York Life IndexFlex Variable Annuity - FP Series offers some investor protections:

- **Death Benefit:** This annuity offers a guaranteed death benefit, which means if you pass away before annuitization (taking income payments), your beneficiary will receive a minimum amount, typically your initial premium or account value.
- **Fixed Account:** The Fixed Account option within the IndexFlex Annuity offers a guaranteed minimum return on your principal. This protects you from losing money in that specific account option.
- **Surrender Charges:** While surrender charges can limit your access to your money early on, they can also discourage people from withdrawing during market downturns, potentially protecting your long-term growth.
- **Free Look Period:** You typically have a 10-day free look period after purchasing the annuity to review the contract and cancel it if you're unsatisfied.

Individual answers

1. Student A: MD RASHEDUL ISLAM RASEL

Understanding various risk factors that could impact the performance of an open-ended, low-cost mutual fund (Vanguard 500 Index Fund) that tracks a broad market index is essential. The breakdown based on designated categories is as follows:

1.1. Collateral-Related Risks:

The Vanguard 500 Index Fund, which mirrors the S&P 500, has minimal credit risks associated with individual company defaults and counterparties in derivative transactions. The fund's broad diversification helps to reduce issuer credit risk. Limited coverage is provided by organisations like the Securities Investor Protection Corporation (SIPC) in case of brokerage insolvency, but not against market losses. Regulatory bodies such as the SEC oversee and ensure investor protection and transparency. Fund managers actively manage credit risks through

thorough research and risk mitigation strategies. Investors should consider these risks in addition to the fund's exposure to the market and align their investment decisions with their risk tolerance and financial goals.

1.2. Statistical-Related Risks:

The Vanguard 500 Index Fund is an open-ended, low-cost mutual fund that tracks the S&P 500, providing broad market exposure. Diversification reduces risk, but a high correlation among its equities increases systemic risk, limiting diversification benefits and hedging opportunities. Investors should inquire about the fund's liquidity during market declines and its plan for managing downside risk. Evaluating historical performance and risk mitigation measures can help assess the fund's resilience in challenging market conditions, aiding in making informed investment decisions.

1.3. Magnifying Risk:

The Vanguard 500 Index Fund is a popular mutual fund that aims to protect investors from losses while offering the potential for gains. It tracks the S&P 500, exposing investors to various large U.S. companies and reducing the risk associated with individual companies. The fund keeps costs low, avoids complex strategies, and does not use borrowed money, which helps shield investors from big losses when the market goes down. By diversifying across many different companies, the fund also reduces the risk of losses. The fund's low costs help maximise potential gains, allowing investors to capitalise on the market's long-term growth. In summary, the Vanguard 500 Index Fund offers a cost-effective way for investors to participate in market growth while managing risks associated with complex investment strategies.

The Vanguard 500 Index Fund has challenges in efficiently managing large amounts of money, not offering real-time trading, and providing limited information about its holdings. These issues can affect how investors experience the fund, manage liquidity, and make investment decisions.

2. Student B: Le Nguyen Dang Khoa

As a financial advisor reviewing the SPDR S&P 500 ETF Trust (SPY), it's important to consider various risk factors associated with this ETF:

2.1. Collateral-Related Risks:

The SPDR S&P 500 ETF Trust primarily invests in companies that mirror the performance of the S&P 500, which include significant large-cap U.S. stocks such as Microsoft Corp and Apple Inc, with significant shares held.

Top Holdings

Fund Top Holdings as of May 17 2024

Name	Shares Held	Weight
MICROSOFT CORP	87,694,313	7.00%
APPLE INC	171,310,876	6.18%
NVIDIA CORP	29,151,095	5.12%
AMAZON.COM INC	107,881,194	3.79%
META PLATFORMS INC CLASS A	25,965,063	2.33%
ALPHABET INC CL A	69,549,400	2.33%
ALPHABET INC CL C	58,228,455	1.96%
BERKSHIRE HATHAWAY INC CL B	21,466,637	1.70%
BROADCOM INC	5,194,781	1.38%
ELI LILLY + CO	9,408,224	1.38%

Index Top Holdings as of May 17 2024

Name	Weight
Microsoft Corporation	7.01%
Apple Inc.	6.19%
NVIDIA Corporation	5.13%
Amazon.com Inc.	3.79%
Meta Platforms Inc Class A	2.33%
Alphabet Inc. Class A	2.33%
Alphabet Inc. Class C	1.97%
Berkshire Hathaway Inc. Class B	1.70%
Eli Lilly and Company	1.38%
Broadcom Inc.	1.38%

(Figure from Top Holdings SPDR® S&P 500® ETF Trust[3])

The diversification from this holding helped the ETF eliminate the credit risk , there remains an inherent credit risk related to the financial health of its underlying assets. Dividends paid by the SPY are sourced from the earnings of these underlying assets and are distributed to shareholders, linking the health of these dividends directly to the performance of included companies.

2.2. Statistical-Related Risks:

The statistics of SPY have a medium to high correlation with the equities comprising the S&P 500 index, which it aims to track. This high correlation underscores its design to closely mimic the index's movements.
However, this also exposes the ETF to systemic market changes that impact the broader market. While SPY's diversification may cause the effects of volatility on individual securities, its overall volatility is still reflective of broader market dynamics.
Historical trends suggest that market-wide volatility influences SPY, although its diversified approach helps alleviate some negative impacts.

2.3. Magnifying Risk:

The SPY does not engage in strategies involving leverage or non-linear investment tactics. It is ideal to replicate the performance of the S&P 500 Index closely, ensuring that returns align closely with those of the index. Unlike products such as indexed annuities that Xing describes, SPY does not incorporate leverage or mechanisms designed to offer downside protection, which places an emphasis on the inherent market risks associated with equity investment in the index. While the Vanguard 500 Index Fund has challenges in efficiently managing large amounts of money, not offering real-time trading, and providing limited information about its holdings. These issues can affect how investors experience the fund, manage liquidity, and make investment decisions.

3. Student C: Xing Xin

3.1. Collateral-Related Risks

The New York Life IndexFlex Variable Annuity - FP Series (NYLIFVA) has reduced credit risk compared to some other investments, but it's not entirely without it.

- **Issuer Risk:** While New York Life Insurance and Annuity Corporation (NYLIAC) is a reputable insurer, there's still a possibility (though unlikely) that they could face financial difficulties. In such a scenario, there's a chance they may not be able to meet their obligations under the annuity contract.
- **Fixed Account Option:** The creditworthiness of the underlying issuers of the bonds held in the Fixed Account option can impact the returns and guarantees offered. If these issuers default, it could affect the value of the Fixed Account.

The NYLIFVA does not have external insurance on its variable investment options or the account value itself. Variable annuities are investment products, not FDIC-insured deposits. And these associations typically protect policyholders in case of insurance company insolvency, but their coverage may not extend to variable annuities.
Since the NYLIFVA has limited credit guarantees, here are some questions you should ask New York Life:

- **NYLIAC's Financial Strength:** Inquire about NYLIAC's financial ratings from independent agencies like A.M. Best, Moody's, and Standard & Poor's.
- **Fixed Account Underlying Issuers:** Ask about the credit quality of the bond issuers within the Fixed Account option.
- **Alternatives for Stronger Guarantees:** Discuss if there are other annuity products offered by New York Life with stronger external guarantees.

3.2 Statistical-Related Risks

Suppose the equities in the index have medium to high correlation. I thinks this indexed annuity is relatively safe, because:

- **Reduced Volatility:** If the underlying equities in the index have medium to high correlation and the chosen Index Account option tracks that index closely, the IndexFlex Annuity might offer reduced volatility compared to directly investing in individual stocks. The index can smooth out some of the ups and downs of the market.
- **Market Risk Remains:** However, the IndexFlex Annuity is still subject to **market risk**. Even with a correlated index, a significant market downturn can negatively impact your investment. The variable investment options within the annuity fluctuate with the market.

My questions about down markets:

For the mutual fund salesperson:

- How do the funds you offer typically perform during down markets? (Historical data and risk ratings)
- Are there any specific share classes that offer downside protection features?

For the ETF salesperson:

- How does the underlying index the ETF tracks typically behave during recessions?
- Does the ETF offer any built-in features to mitigate losses during volatile markets?

3.3 Magnifying risk factors

The index account option tracks a market index for downside protection, but it typically doesn't directly reflect the index's full negative performance. There are mechanisms to limit the downside:

- **Minimum Index Crediting Rate:** The annuity contract might guarantee a minimum crediting rate, even if the index loses value. This provides a floor for your returns.
- **Participation Rate:** This limits how much of the index's gains are credited to your account. While it caps potential losses, it also caps potential gains.

I think there are two points for the upside potential:

- I**ndex Tracking:** If the chosen Index Account option tracks a rising market index, the annuity's value can grow. The participation rate determines how much of that growth is credited to your account.
- **Annual Reset:** Some Index Account options offer an annual reset feature. If the index increases during a year, the new high point becomes the baseline for the next year. This can lock in gains and offer potential for further growth.

The participation rate in an indexed annuity doesn’t function as true leverage, because the participation rate usually has a cap, typically around 100%. This means it won’t amplify equity market gains beyond a certain point. And the participation rate also affects potential losses. A lower participation rate reduces the negative impact of a declining market but also reduces the upside from a rising market.

3.4 Frictional risk factors

The New York Life IndexFlex Variable Annuity - FP Series offers some potential benefits, but there are some limitations caused by liquidity and regulation.

Liquidity:

- Surrender Charges**:** These fees are applied if you withdraw money within a certain period (usually up to 7 years) from buying into the annuity. Early withdrawals can significantly reduce your accessible cash.
- Segment Withdrawals**:** If you withdraw money from an indexed investment option (index-linked account) before the end of its term (segment maturity date), you forfeit any potential growth from that period. This limits your ability to access your money even within the surrender charge period.

Regulation:

- Tax Implications: Early withdrawals (before age 59.5) from the annuity may be subject to income tax and a 10% penalty. This discourages accessing your money for short-term needs.
- State Guaranty Associations: These associations typically protect policyholders in case of insurance company insolvency. However, their coverage may not extend to variable annuities (check with your state's department of insurance). This means your principal might not be fully protected in case of unforeseen circumstances.

3.5 Fallout risks

The New York Life IndexFlex Variable Annuity - FP Series itself doesn't have direct fallout risks in the traditional sense. However, the IndexFlex Annuity can be indirectly impacted by fallout events due to its reliance on the financial system. Below are two fallout events could impact the indexFelx Annuity:

- **Market Downturn:** The variable investment options within the IndexFlex Annuity are subject to market risk. A significant fallout event could trigger a major market crash, leading to substantial losses in the value of your investment.
- **Regulation Changes**: In response to a major economic crisis, governments might implement new regulations on financial products, including variable annuities. These changes could impact the terms of your annuity contract or limit its benefits.

Group Report

1. Mutual Fund

Investing in the Vanguard 500 Index Mutual Fund, investors may encounter capital flows, trading, and transparency challenges.

1.1. Large Capital Flows:

Substantial capital flows in and out of the fund can give rise to several issues. When large amounts of money are invested or withdrawn, the fund must buy or sell any securities to align with the index. This can impact market prices, potentially increasing costs for the fund. Moreover, significant inflows might result in the fund temporarily holding a higher cash position, which can dilute returns since cash typically earns less than equities. Conversely, substantial outflows may compel the fund to sell holdings at unfavourable times, potentially locking in losses or missing out on gains

1.2. Lack of Intraday Trading:

The absence of intraday trading for the fund entails trading at the end-of-day Net Asset Value (NAV). This limits investors' ability to react to market movements in real time, hindering their capacity to capitalise on intraday price fluctuations or promptly respond to market news. Orders placed during the day are executed at the end of the NAV, which can pose challenges in volatile markets where prices can change significantly within a day. In contrast, exchange-traded Funds (ETFs) tracking the same index, such as the Vanguard S&P 500 ETF (VOO), allow for intraday trading, making them more attractive to investors desiring greater trading flexibility.

1.3. Limited Transparency of Holdings:

Mutual funds typically disclose their holdings every quarter, leading to limited transparency of holdings. By the time holdings are disclosed, the information needs to be updated, which can disadvantage investors who prefer real-time knowledge of the securities they hold at any given time. This lack of real-time transparency can also raise concerns about the fund's management strategies, especially in times of market volatility. ETFs often provide daily disclosure of their holdings, offering greater transparency and making them more appealing to investors who value frequent updates on the portfolio's composition.

2. Exchange-Traded Fund (ETF): The SPDR S&P 500 ETF Trust (SPY)

2.1. Market Volatility Affecting Performance:

Because the raw idea is tracking an index so market volatility plays a significant role in the performance, SPY is directly influenced by the fluctuations in the stock prices of the 500 largest publicly traded companies in the United States. In the bull-market with the stock prices increase would lead to the SPY price increase but in the bad scenario where a crisis or bear-market the SPY also badly affected. So the investment in SPY should be prepared for these fluctuations and consider their risk tolerance when investing in this ETF.

2.2. Fees Associated with Management and Exchanges:

SPY is known for its low expense ratio, which is one of its key advantages over actively managed funds. The current expense ratio for SPY is approximately 0.09% as I mentioned earlier in Section 2. However, there are other fees to consider beyond the management fee. Investors also face transaction fees when buying or selling shares of SPY on an exchange. These fees can include brokerage commissions and the bid-ask spread, which is the difference between the price at which buyers are willing to purchase the ETF and the price at which sellers are willing to sell it. While these costs are generally low for highly liquid ETFs like SPY in compare with other funds:

Bid/Ask Spread Comparison

	SPY	IVV	VOO
Share Price	523.07	525.73	480.70
Bid/Ask Spread ($)	0.011	0.022	0.020
Bid/Ask Spread (bps)	0.21	0.42	0.45

Source: Bloomberg Finance L.P., State Street Global Advisors, as of 03/31/2024.
Bid/ask spreads are the average of the month of March 2024.

12-Month Average Bid/Ask Volume Comparison

	SPY	IVV	VOO
Bid Size	2,807	1,754	1,718
Ask Size	2,785	1,643	1,655

Source: New York Stock Exchange, State Street Global Advisors, as of 03/31/2024.

They can add up, especially for investors who trade frequently. Additionally, there may be other administrative fees or taxes depending on the investor's country of residence and the regulations applicable in their market.

2.3. Dividends Impacting the Fund:

Dividends also play a crucial role in the total return of SPY. The ETF distributes dividends to its shareholders based on the dividends received from the underlying stocks in the S&P 500 Index as described earlier. These dividend payments can provide a steady income stream for investors and contribute to the overall performance of the fund. However, the timing and amount of dividend payments can vary depending on the performance of the underlying companies and their dividend policies.
For example, during periods of economic stress, companies might not maintain their maintenance, which would lead to reduction or suspend their dividend payments, which would decrease the dividends distributed by SPY. Conversely, in a robust economic environment, companies may increase their dividends, benefiting SPY investors. It is also important to note that dividends are typically paid out quarterly and are subject to taxation, which can impact the net returns for investors.

In conclusion, the SPDR S&P 500 ETF Trust (SPY) provides broad exposure to the U.S. equity market, offering advantages such as low fees and the potential for dividend income. However, it is important to remember that, like stocks, ETFs have investment risks. They fluctuate in market value and may trade at prices above or below the ETF's net asset value. We must be mindful of market volatility, transaction fees, and the variability of dividends. Additionally, concern about the brokerage commissions and ETF expenses can reduce returns.

3. Indexed Annuity: New York Life IndexFlex Variable Annuity

3.1. Penalties for early withdrawal

- **Discourages Short-Term Investing:** The IndexFlex Annuity is designed for long-term goals like retirement. Early withdrawal penalties make it a poor choice for short-term savings needs. You might lose a portion of your principal due to the fees.
- **Reduced Return Potential:** If you're forced to withdraw money early due to a financial hardship, the surrender charges eat into your investment returns. This can significantly impact your overall financial goals.

3.2. The lack of control of the participation rate

- **Limited Upside Potential:** The participation rate determines what percentage of the index's gains are credited to your account. A low participation rate offers more protection from market downturns but also limits your potential for growth during bull markets. You don't have control over this rate, and it can be capped, further limiting upside.

- **Difficult to Optimize Strategy:** Since you can't adjust the participation rate, it's challenging to tailor the annuity's risk-reward profile to your specific risk tolerance. You might end up with a more conservative or aggressive allocation than you prefer.

3.3. The limited transparency of fees

- **Hidden Costs:** The IndexFlex Annuity has several fees, including premium charges, sales charges, underlying fund expenses, contract fees, and surrender charges. While a prospectus outlines the fees, it can be complex and difficult to understand the total impact on your investment.
- **Difficulty in Cost Comparison:** The various fees associated with the annuity can make it challenging to compare it directly with other investment options that might have a simpler fee structure. This can make it difficult to assess the true cost of ownership.

Regulation: Describing the crisis from the reading

Student B: Le Nguyen Dang Khoa

Reading B: "There is some discrepancy between physical and synthetic ETFs". (A sample article)

The Exchange-Traded Funds (ETFs) have become essential in every investment strategy. Rely on the assets they track or the fund management most ETF use derivatives like swaps to replicate an index's performance. This distinction has raise a debate among investors and regulators, as synthetic ETFs present both unique advantages and potential risks

- Understanding Synthetic ETFs:

Synthetic ETFs are designed to copy the performance of an index without directly holding the securities in that index. Instead, they enter into a swap contract with another financial entity. This approach can lead to **enhanced tax efficiency and lower transaction costs**, making them attractive for certain markets, particularly commodities and international indices where direct investment is either costly or cumbersome.

- The Crisis of Perception and Performance:

Despite their efficiencies, synthetic ETFs have faced skepticism due to their complex nature and the perceived higher risk associated with their reliance on derivatives and counterparties. The crisis of confidence stems from a lack of understanding and the fear that these instruments could be vulnerable to market abnormalities or counterparty defaults, especially during financial downturns.

- Comparative Analysis: Synthetic vs. Physical ETFs:

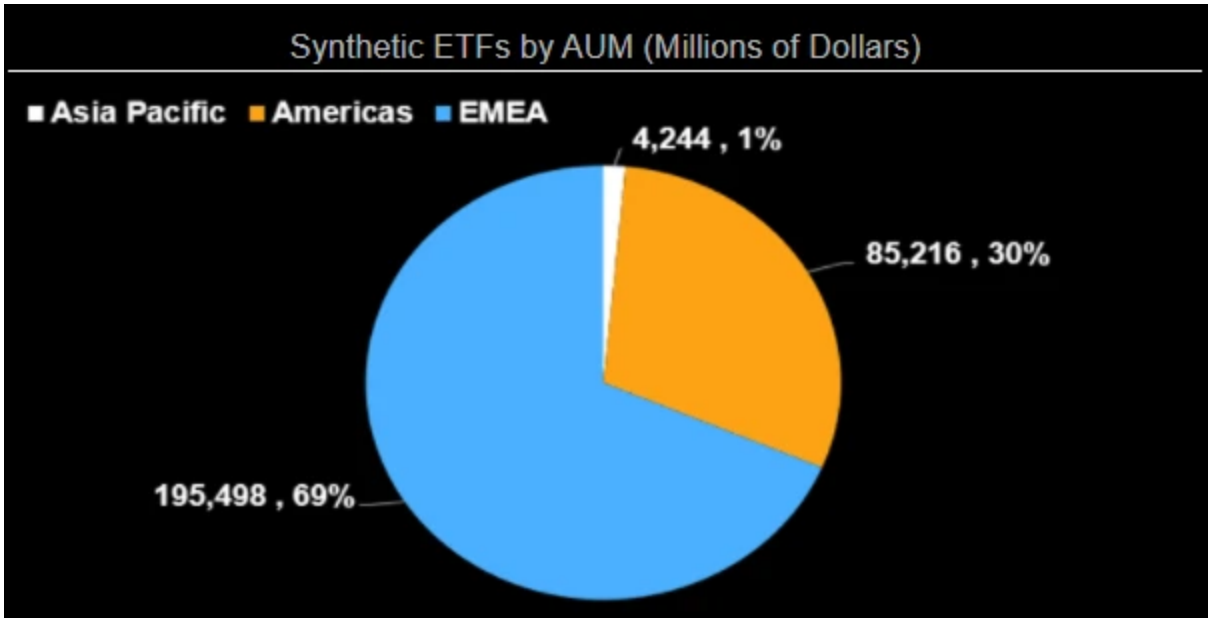
Data from Bloomberg shows that synthetic ETFs, like the iShares S&P 500 Swap UCITS ETF, have outperformed their physical counterparts in certain instances.

For example: Over a one-year period, the iShares synthetic version outperformed by 47 basis points, primarily due to lower costs and tax efficiencies. However, not all synthetic ETFs perform equally, and their success can vary based on the underlying assets, swap arrangements, and market conditions.

(Source: Bloomberg)

- Regional Preferences and Tax Considerations:

The utilization of synthetic ETFs varies significantly across regions. In EMEA, synthetic ETFs are more prevalent, making up about 67% of the market, largely due to favorable tax regimes under UCITS. In contrast, the Americas and Asia-Pacific have been slower to adopt, with a focus mainly on commodities and less on broad market indices.



(Source: Bloomberg)

- Proposed Regulations:

To address the crisis and improve the quality of synthetic ETFs, several regulatory improve could be release such as enhanced transparency, standardization of practices, strengthened counterparty Risk Assessment

- Conclusion:

Synthetic ETFs offer unique advantages that can be matched effectively if accompanied by informed regulation and investor knowledge.Because they help optimize opportunity for tax-efficient, cost-effective investment strategies, particularly in complex markets.
-> As such, re-evaluating the regulatory approach to synthetic ETFs could enhance their acceptance and utility in the global financial landscape.

Student A: MD RASHEDUL ISLAM RASEL

In 2003, there was a mutual fund crisis.

The article discusses the impact of Japan's "Big Bang" deregulation on its financial landscape, drawing parallels with similar movements in the US and Britain (Toya, 2006). It examines how Japan aimed to revitalise its economy and compete globally, mirroring strategies adopted by other major financial hubs. The deregulation of financial markets, akin to the "Big Bang" in Britain, was envisioned as a transformative event, revitalizing Japan's economic universe.

However, Japan's journey toward deregulation was not without challenges. Widespread corruption and financial scandals in the 1980s and 1990s necessitated reforms, albeit with the acknowledgement that reform alone cannot eliminate all problems. The article highlights the deregulation's impact on the mutual fund market, with changes aimed at increasing

transparency and expanding investment options. Yet, it points out persisting issues, such as the historical lack of transparency in Japanese investment trusts and the challenge of ensuring investor trust in the absence of adequate information.

Furthermore, the article explores the cultural dynamics influencing white-collar crime in Japan. It contrasts Japanese business norms, emphasising conformity and hierarchical structures, with the more whistleblower-friendly environment in the US. This cultural backdrop, combined with the gradual pace of deregulation in Japan, shapes the landscape for financial fraud. While deregulation opens avenues for abuse, the cultural emphasis on relational investing may mitigate some risks, although challenges persist in detecting and addressing fraud.

Drawing on historical events like the 2003 mutual fund crisis, the article underscores the importance of regulatory vigilance and responsiveness. It compares regulatory responses in the US and Britain to crises, highlighting the need for Japan to adapt swiftly to emerging challenges to avoid economic collapse. The article suggests that while Japan's unique cultural and regulatory context may influence the nature of financial fraud, proactive measures are essential to safeguard the integrity of the financial system.

Student C: Xing Xin

1. Crisis

When a broker sells a product that spans multiple regulatory schemes the oversight becomes very complicated.

2. Proposed Regulations

A broker's core obligation is to ensure their recommendations are suitable for the client's individual circumstances and financial goals. This requires a comprehensive understanding of the product, the client's profile, and potentially applicable regulations.

FINRA Rule 2111 applies to recommendations of securities offered by FINRA-registered broker-dealers. It outlines three main obligations for brokers:

a) Reasonable-Basis Suitability: The broker must have a reasonable basis to believe the recommended security or investment strategy is suitable for at least some investors. This involves understanding the product's risks and rewards.
b) Customer-Specific Suitability: The recommendation must be suitable for the particular client's investment profile. This includes considering them:
    - Investment objectives (growth, income, etc.)
    - Risk tolerance (how much risk they're comfortable with)
    - Financial situation (income, assets, liabilities)
    - Investment experience and knowledge

c) Quantitative Suitability: The broker should consider the client's investment horizon (timeframe), liquidity needs (access to cash), and other relevant factors to ensure the investment aligns with their financial capabilities.

Variable annuities have investment features and may be subject to FINRA's suitability rule in some cases. Here are some additional considerations for brokers. First, the broker must thoroughly understand the specific variable annuity's features, underlying investments, fees, and risks.Second, the broker should discuss alternative investment options that might better suit the client's needs, especially if the variable annuity has complex features or high fees. And variable annuities can have tax implications for withdrawals. The broker should ensure the client understands these potential tax consequences.

The regulatory treatment of EIAs can be complex. Below is how a broker's obligations might differ:

- FINRA vs. State Regulation: Depending on the specific features of the EIA, it might be considered a security subject to FINRA's suitability rule or solely an insurance product regulated by individual states. The broker needs to determine the appropriate regulatory framework.
- Focus on Insurance Aspects: If the EIA is primarily viewed as insurance, the broker's focus might shift towards explaining the death benefit, surrender charges, and other insurance-related features.
- Disclosure Requirements: State regulations might have specific disclosure requirements for EIAs. The broker must ensure compliance with these requirements to adequately inform the client.

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